PhD - The day after

Some possible career paths after a PhD



David Pichardie - October 19th, 2026

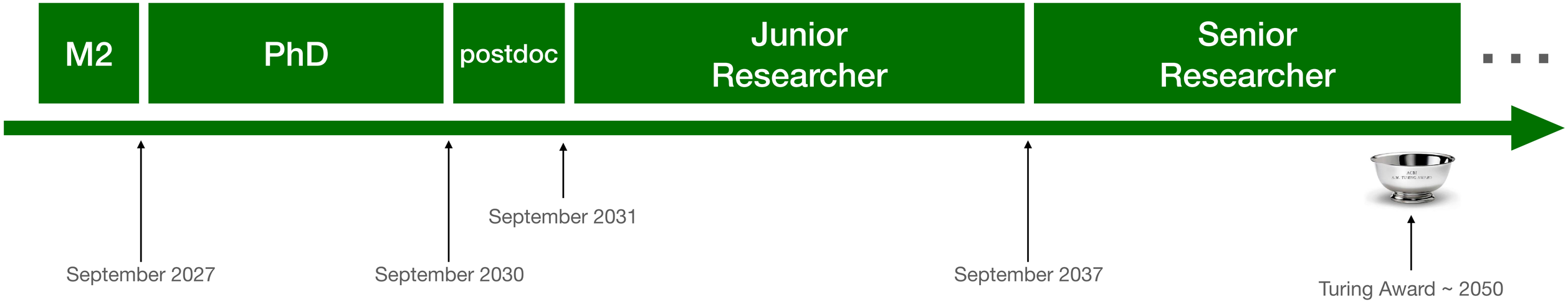
Why bother now?

- To make you want to do a great PhD
- To make you want to do a great Master 2
- To make you want to choose your Master's internship carefully



- And because a little anticipation never hurts…

THE Roadmap


M2
PhD
postdoc
Junior Researcher
Senior Researcher
September 2027
September 2030
September 2031
September 2037
Turing Award ~ 2050


Wrong!

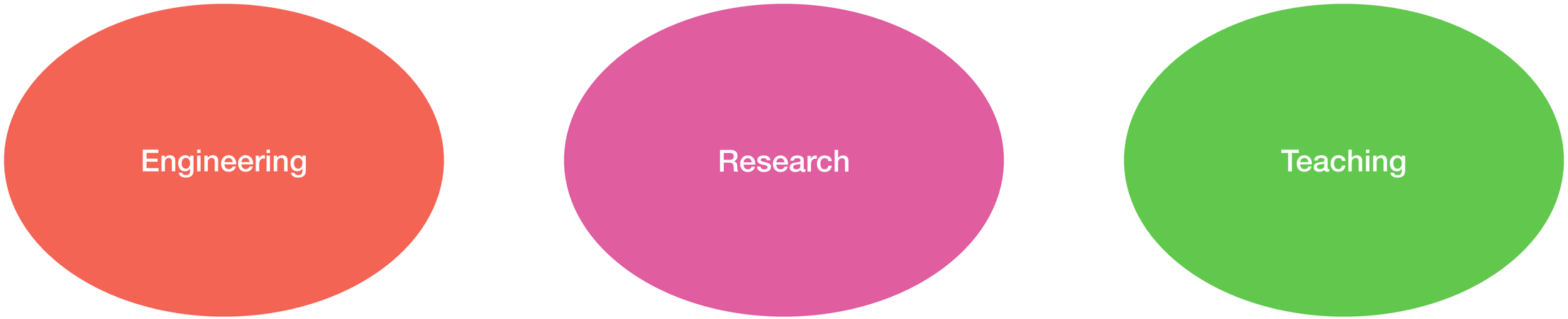
There is no ONE roadmap

- Many careers are possible
- Each of you has different skills, wishes and family constraints
- Don’t let this talk decide for you

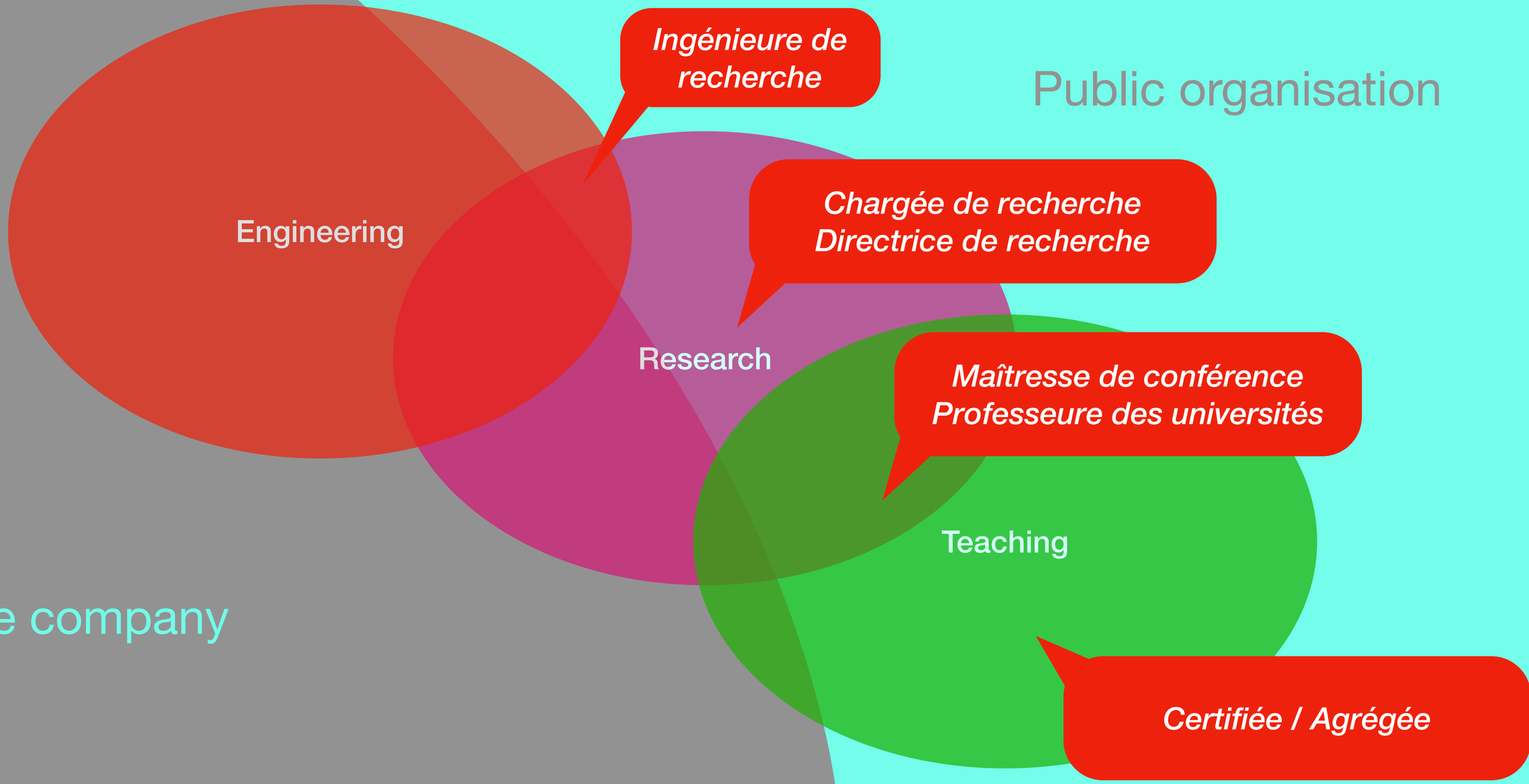


- But take the time to learn about
  - the different paths
  - and the specific **rules** of the game

The main categories of careers

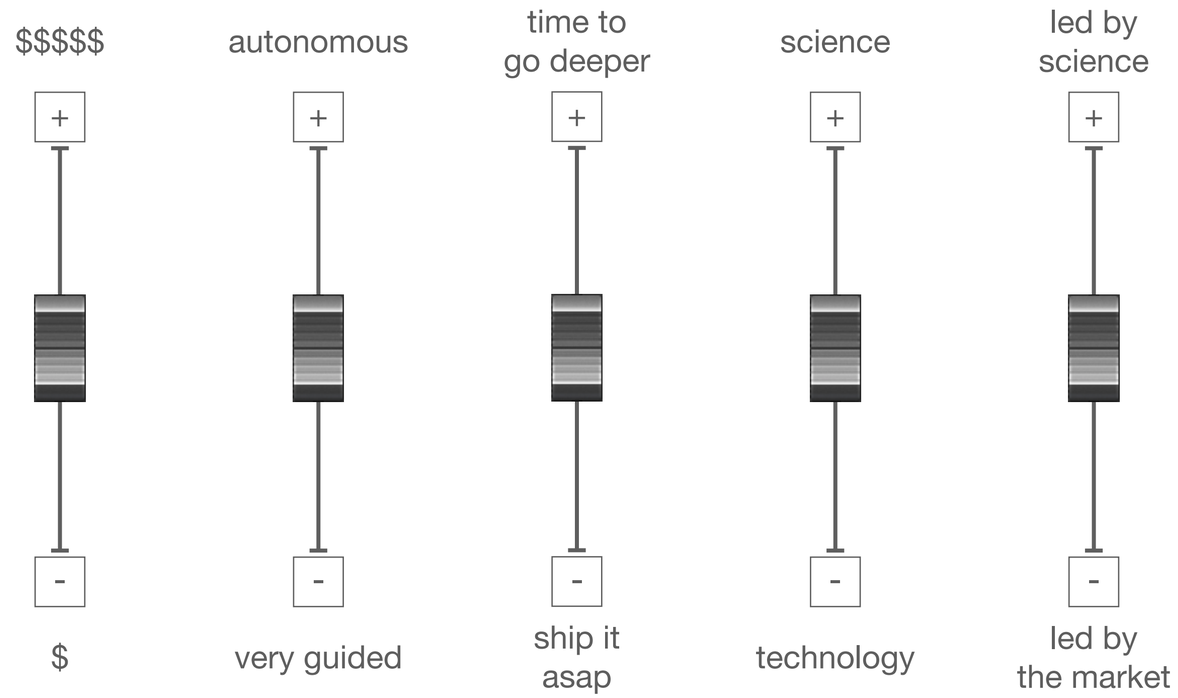


The main categories of careers



Choosing your career

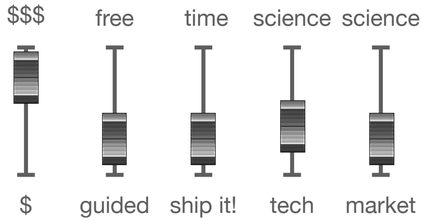
… in a multi-dimensional space



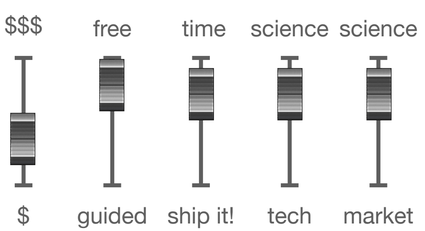
Practical exercises

Put the sliders where you expect them to be

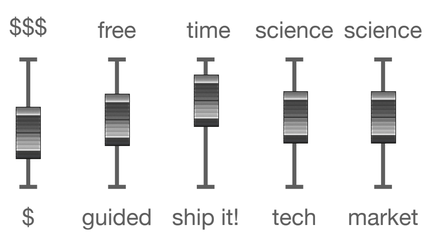
Engineer



Researcher



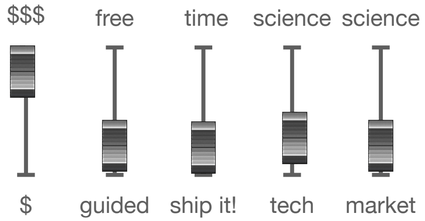
Teacher



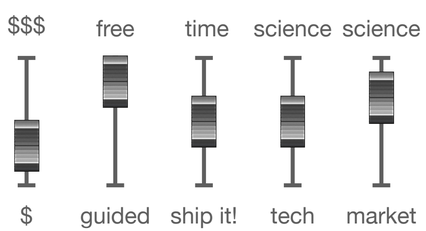
Practical exercises

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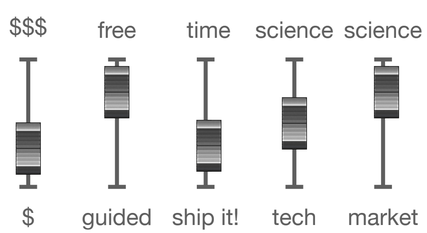
Engineer



Researcher



Teacher



Academia

Chercheurs & enseignants-chercheurs



- *Chercheurs*: INRIA, CNRS, CEA
- *Enseignants-chercheurs*: universities, *grandes écoles*
- *You will spend the next 4 years (including M2) with them!*
  - take the time to observe them
  - the reality: the number of permanent positions has decreased
  - beware: (French) people always complain about their work!

but not as much in Computer Science as in other domains

Don't listen too much to the grumps!



Industry

- Large / mid-size companies / startups
- Each company has its own culture
- The level of science may depend a lot on the company/team
- But *Time is money*
- French industry does not always value PhD degrees, but at the same time it needs deep expertise on some specific topics.



Teaching computer science in high school

- Many opportunities over the next decades
  - CAPES created in 2020
  - Agrégation created in 2022
- School curricula will contain more and more computer science
- A PhD is not required, but you will never regret this unique opportunity to focus on deep science



Step 1: choose your career

Step 2: get the job!

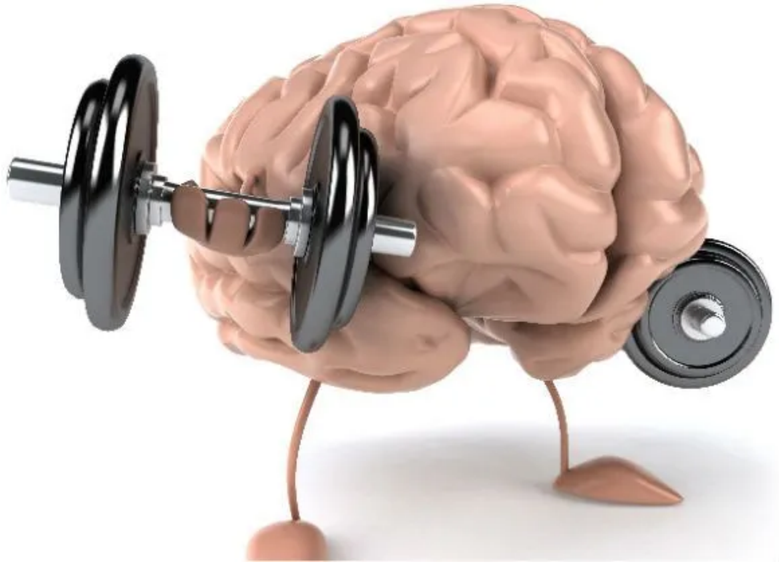
My advice

- I will give you some advice
- Several are inherited from a previous version of this talk (by Luc Bougé)
- Take them with a *pinch of salt* and discuss them with other colleagues when the time comes



Be competent

- We expect your PhD to make you an expert in your (narrow) domain
  - don’t forget that
  - and enjoy this *bubble of science*
  - go deep
  - read
  - experiment
  - meet the gods of your domain



Improve your communication skills

- After (and a bit during) your PhD, you will have to explain your area of expertise to non-experts
- You can practice during your PhD
- Communication can be oral or written
  - written: you will have to write research proposals
  - oral: you will have to go through job interviews

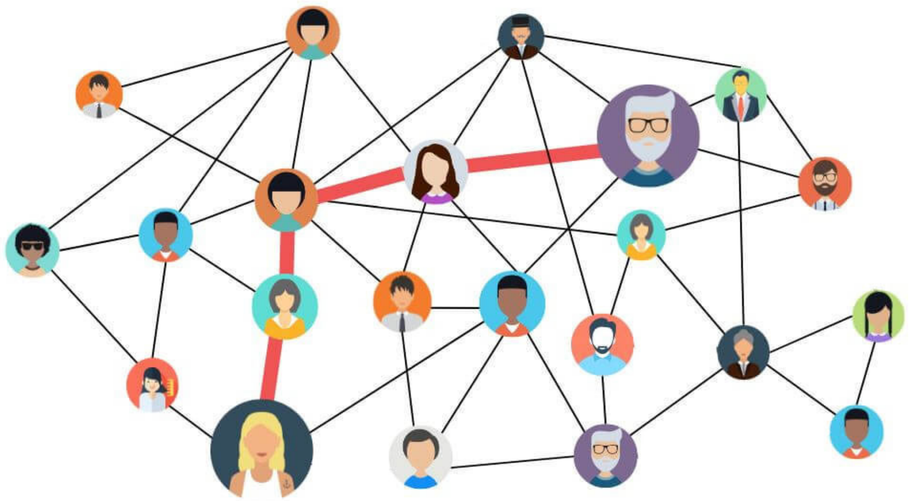


and of course research papers

and of course research talks

Build a network

- Don’t hide in your PhD office for 3 years!
- Meet as many people as you can
- Listen to their own experience
- Don’t talk only to PhD students and postdocs
- Also talk to senior people, and put yourself in the shoes of those who will hire you later



Seminars

Conferences

Summer schools

Cafeteria

Fête de la science

Publish

- Full research positions require a first-class publication record (PhD+postdoc)
- Teaching positions at university are less demanding
  - but still, you will be mainly evaluated on your research, not on your teaching skills
- Your advisor will give you the keys
  - find one who best fits your career plans
- You can also get a PhD with just *honest* publications, if you plan to join industry or high school teaching afterwards

Keep in mind that the USA ≠ France



Not all publications are equal

- Conference papers: the main venue in most of computer science
  - full papers (≈10-25 pages), peer-reviewed, selective
  - short papers, tool papers, posters: lighter and less weighted
- Workshops: good to start, to get feedback and meet people, but often little weight in an evaluation
- Journals: slower, more complete, more valued in some areas
- Others: artifacts, preprints (arXiv, HAL), thesis
- How to judge a venue?
  - CORE ranking https://portal.core.edu.au/conf-ranks/: conferences ranked A*, A, B, C
  - Scimagojr ranking https://www.scimagojr.com/: journal ranked Q1, Q2, Q3, Q4
- Want to find someone publication record?
  - Go to https://dblp.org/



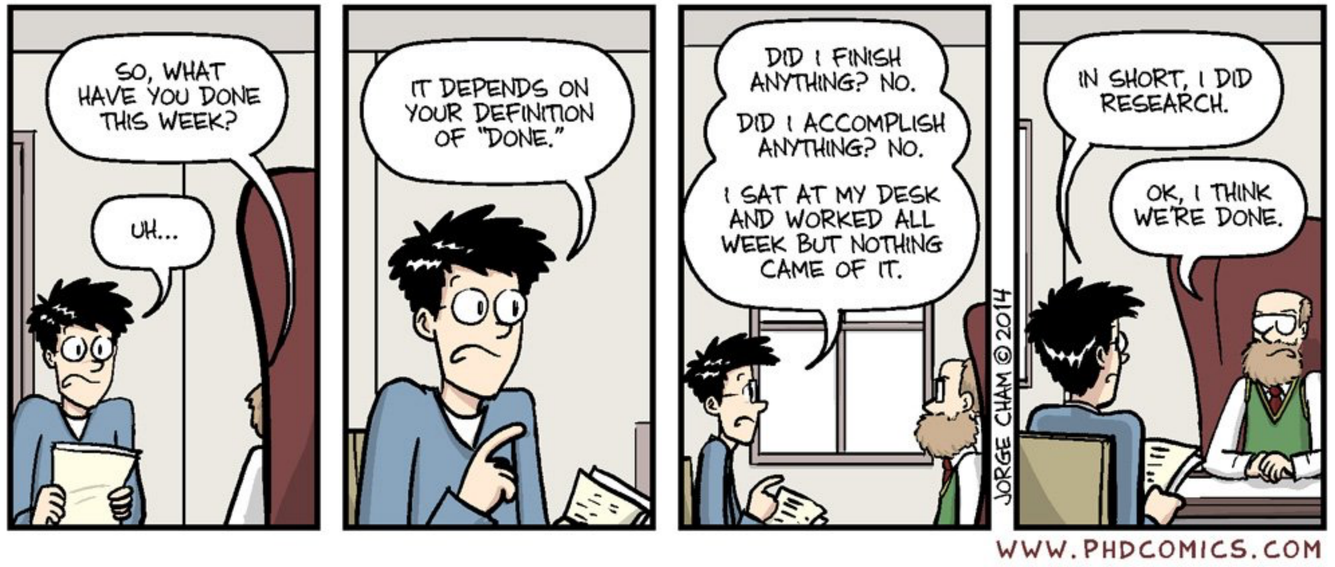
Anticipation

- Applying for a position takes about a year
  - searching for openings
  - meeting the right people, understanding their needs
  - building up your application
    - full CV
    - research proposal
      - short-term / mid-term / blue sky
    - teaching experience
    - reference letters



Get the most out of your advisor

- You are responsible for reaching out to them
  - plan regular meetings
  - prepare well to have an effective meeting
  - take notes, manage archives
- Keep in mind
    - you are (mostly) mono-threaded
    - They are multi-threaded



You shall

- Learn
- Communicate
- Meet people
- Publish
- Anticipate
- Listen to your advisor

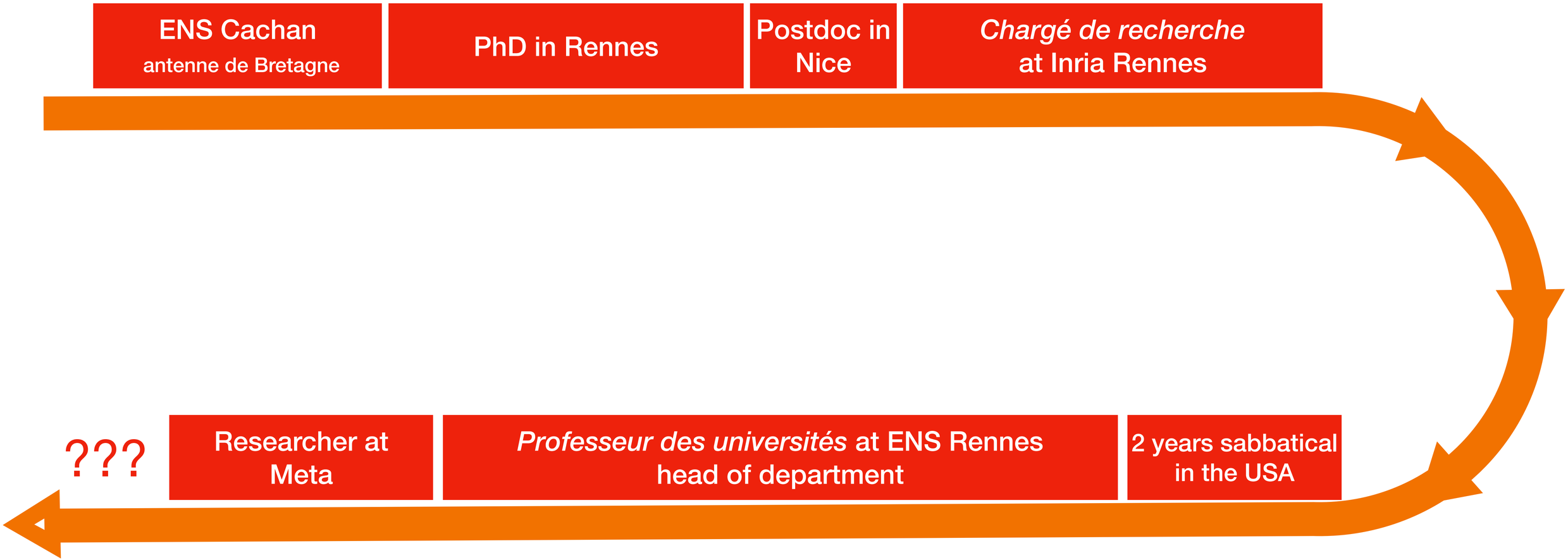


Don’t be afraid of planning

- It’s not easy for anyone
- The future
  - may be frightening
  - but you have plenty of opportunities ahead of you
  - just find them!



My personal journey


ENS Cachan
antenne de Bretagne
PhD in Rennes
Postdoc in Nice
Chargé de recherche at Inria Rennes
2 years sabbatical in the USA
Professeur des universités at ENS Rennes
head of department
Researcher at Meta
???

My personal thoughts

- I could have failed many times
  - especially because of self-censorship
  - every time, advice from colleagues helped me get unstuck
- I’ve been lucky enough to do several different things
  - my tastes have changed over time
  - we have to make our own luck!



Good luck